AI-Assisted Systematic Reviews and Meta-Analysis

Day 1: From Query to Corpus

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instats Seminar

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Day 1: From Query to Corpus

Interactive App and Repository

- **Streamlit App URL:** Click here to open the Interactive Streamlit App
- **No Installation:** Runs entirely in the browser via Streamlit — no Python or R required.
- **Navigation:** Use the sidebar to switch between the three seminar days.
- **Step-by-Step:** Each page walks through the core extraction, screening, and synthesis computations for that day.
- **Consistency:** Results match the companion Python and R scripts available on the seminar GitHub page.
- **GitHub Page URL:** Click here to open the seminar GitHub page

The Crisis of Evidence Synthesis

Systematic reviews remain one of the most time-intensive forms of research. Across the health and life sciences, the social sciences, the humanities, and engineering, researchers are expected to map rapidly evolving fields and evaluate interventions.

The traditional workflow involves:

- Manual database searches
- Screening of thousands of titles and abstracts by two independent reviewers
- Painstaking extraction of data from full-text PDFs

This workflow is characterized by massive time investments, significant human error, and a growing mismatch between the pace of publication and the pace of synthesis.

AI as a Core Component of Research Design

The methodological consequence of this mismatch is clear: if evidence synthesis is to remain feasible and rigorous in an era of information abundance, the integration of computational and AI-assisted tools must be treated as a **core component of research design** rather than an optional convenience.

Funding bodies, journals, and methodological organizations are increasingly acknowledging the need for AI-assisted approaches, while simultaneously calling for transparency, validation, and the preservation of human oversight (Flemyng et al., 2025).

A Methodologically Grounded Approach

The scientific value of AI-assisted synthesis depends less on the mere act of automation than on:

- Transparent query formulation
- Rigorous bias mitigation
- Structured data extraction
- Adherence to established reporting guidelines (e.g., PRISMA 2020)

AI is best understood and taught as a powerful assistant within a rigorous methodological framework, requiring explicit validation and human verification at each stage.

The Shift from Siloed Databases to Open APIs

The data collection phase of evidence synthesis is undergoing a transformation. The rise of open bibliographic metadata catalogs allows researchers to bypass proprietary, siloed databases.

API Source	Coverage Focus	Strengths
OpenAlex	Universal (cross-disciplinary)	Massive scale, concept tagging, citation graphs
Semantic Scholar	Computer Science, Biomedicine	High-quality abstracts, AI-driven extraction
Crossref	Universal (DOI-based)	Official metadata, direct publisher links

These APIs allow researchers to construct comprehensive corpora **programmatically and reproducibly** (Delgado-Quirós and Ortega, 2024; Culbert et al., 2025).

Corpus Construction is Not Value-Free

In adjacent work on text analysis, Grimmer, Roberts, and Stewart (2022) emphasize that data construction is inseparable from theory, representation, and validation; there is no value-free corpus construction.

This is highly applicable to AI-assisted systematic reviews. An AI-assisted review is only as good as the corpus it operates on. The researcher does not encounter a neutral, perfect set of literature, but rather a dataset shaped by:

- Indexing practices
- API limitations
- The precision of search query formulation

Query Formulation: The PICO Framework

A robust systematic review begins with a substantive question translated into a structured search strategy. The most common framework is PICO (widely used in Health Sciences, but adaptable elsewhere):

- **P**opulation: Who is being studied? (e.g., low-SES patients)
- **I**ntervention: What is the action? (e.g., telemedicine)
- **C**omparator: What is the alternative? (e.g., standard care)
- **O**utcome: What is measured? (e.g., HbA1c levels)

In the Social Sciences, frameworks like SPIDER (Sample, Phenomenon of Interest, Design, Evaluation, Research type) are often used.

Boolean Logic and API Queries

Translating a conceptual framework into an API query requires Boolean logic and field tags. A typical query structure:

```
("health inequalities" OR "socioeconomic disparities")
AND
("diabetes care" OR "chronic disease management")
```

When querying OpenAlex or Semantic Scholar programmatically, these strings are passed to specific endpoint parameters (e.g., searching within title and abstract only, restricting by publication year, filtering out non-journal articles).

The Deduplication Challenge

When aggregating records from multiple API queries or multiple sources, duplication is inevitable. A single paper may appear as a preprint, a conference proceeding, and a final journal article.

Programmatic deduplication relies on a two-pass logic:

1. **Exact Match:** Compare normalized DOIs (Digital Object Identifiers). If DOIs match, keep the most complete record.
2. **Fuzzy Match:** Compare normalized titles (lowercase, punctuation removed) combined with publication year. If titles are highly similar (e.g., Levenshtein distance) and years match, merge the records.

Reporting the Search Strategy (PRISMA-S)

Reproducibility in corpus construction requires adherence to reporting standards like PRISMA-S (Search extension).

A rigorous workflow must log:

- The exact Boolean strings used
- The specific APIs queried (and their versions/dates)
- The number of records retrieved per query
- The number of duplicates removed
- The final size of the deduplicated corpus

The Streamlit app automatically generates this log during the Day 1 workflow.

Discipline-Spanning Examples

This seminar uses four case studies that thread consistently through all three days. Participants can follow the full systematic review lifecycle in a domain close to their own:

Discipline	**Topic**	**Day 1 Focus**
Health Sciences	Health Inequalities in Chronic Disease	Query OpenAlex
Social Sciences	Universal Basic Income (UBI) Outcomes	Query Semantic Scholar
Science/Eng	Microplastic Pollution in Aquatic Environments	Query OpenAlex/Crossref
Management	CSR and Firm Financial Performance	Query OpenAlex

Hands-On: Generating the Raw Corpus

In the second half of today's session, we move to the Streamlit app.

Guided Examples: We will run the pre-configured API queries for the four case studies. We will observe the raw JSON responses, the flattening of nested metadata, and the deduplication process, resulting in a clean CSV corpus.

Bring Your Own Data (BYOD): You will formulate your own Boolean query, select an API (OpenAlex or Semantic Scholar), and run the collection live. You will download your deduplicated corpus and the associated query log, ready for Day 2.

References

- Boudourides, M. (2026). AI-Assisted Systematic Reviews and Meta-Analysis.
- Culbert et al. (2025). Open bibliographic APIs and the infrastructure of retrieval.
- Delgado-Quirós, L., & Ortega, J. L. (2024). OpenAlex as a comprehensive bibliographic database.
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- Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). *Text as Data*. Princeton University Press.

Questions and Discussion

Thank you!

Questions?

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